PETROLEUM PRODUCTION ENGINEERING

MODULE-5

RESERVOIR CONSIDERATIONS

PACKERS, TUBULAR AND FLOW LINES

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Tubulars

- Most oil wells produce reservoir fluids through tubing strings.
- Production tubing is a wellbore tubular used to produce reservoir fluids.
- It forms the conduit for the reservoir fluids to flow from well bore to surface.
- Typically, tubing is run inside a casing or liner but can also be cemented in slim-hole wells as the casing.
- Depending on the type of completion, one or two tubing strings may be used in the well.
- Production tubing is assembled with other completion components to make up the production string.

Tubulars

- Should be compatible with the wellbore geometry, reservoir production characteristics and the reservoir fluids.
- Provide good sealing performance and allow the use of gas expansion to lift oil.
- Gas wells produce gas through tubing strings to reduce liquid loading problems.
- Tubing strings are designed considering tension, collapse, and burst loads under various well-operating conditions to prevent loss of tubing string integrity including mechanical failure and deformation due to excessive stresses and buckling.

Functions of Tubing

- It is essential for well killing, circulation and work over.
- Tubing provides optimized flow channel to produce optimum/maximum efficient rate from a well.
- Tubing is used to isolate casing from effect of high pressure, high temperature and corrosive fluids.
- It facilitates completion, production and control of flow in multiple commingled or parallel completion.
- It is essential for most common artificial lift operations.
- It facilitates the installation of common wire line operated down whole tools.

Packer

- Production packers are an integral part of the completion process of an oil or a gas well.
- Provides a means of sealing the tubing string from the casing thereby preventing communication of fluids.
- Casing used in a well is a permanent component of the complete system, repair/replacement of casing is very complicated and expensive, whereas the packer along with the tubing string is easier to remove and replace.
- Packers are sometimes run when they serve no useful purpose, resulting in unnecessary initial investment and the possibility of high future removal cost.

Uses of Packer

- Routine use of packers should be limited to certain situations:
  1. Protection of casing from pressure (including both well and stimulation pressures) and corrosive fluids
  2. Isolation of casing leaks, squeezed perforations, or multiple producing intervals.
  3. Elimination of inefficient "heading" or "surging".
  4. Some artificial lift installations.
  5. In conjunction with subsurface safety valves.
  6. To hold kill fluids or treating fluids in casing annulus.

Types of Packer

- **Retrievable:** They are run in an integral part of the tubing string. They are set either mechanically or hydraulically and can he released by pulling or rotating the tubing.
- **Permanent:** This type of packer once set, can only be removed destructively by milling. Permanent packers can be set mechanically, hydraulically or electrically (wire line or explosive set) is required.
- **Permanent – retrievable:** Combination

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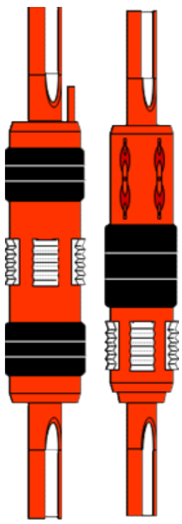
Types of Packer



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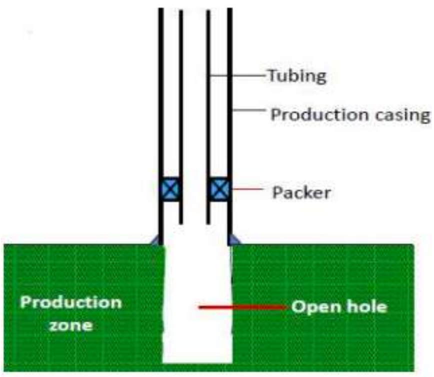
Well Completion Methods

- There are three basic ways to complete a well;
  1. Open-hole
  2. Cased and perforated
  3. Liner completion

Open Hole

- In the open-hole type of completion, casing is set only to the top of or slightly into the completion interval
- In this case the production casing is set on top of or slightly into the pay zone and cemented
- The pay zone is left open and un-cemented



Open Hole: Advantages

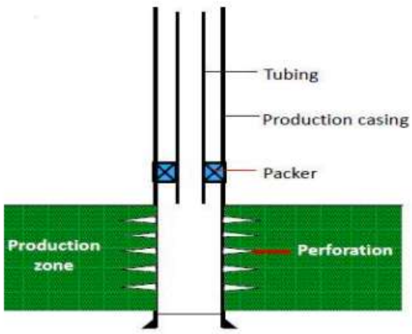
- Casing set at the top of the pay allows for special drilling techniques which minimize formation damage.
- Full-hole diameter available to flow.
- No perforation is required (open-hole perforating is sometimes used in cases of severe wellbore damage).
- Hole -is easily deepened or converted to a liner completion.
- High productivity maintained when gravel packed for sand control.
- Facilitates ultra short radius multiple radial completion.

Open Hole Disadvantages

- No way to regulate fluid flow from or into wellbore.
- Cannot control gas or water production effectively. No selective gas or water shut off job is possible.
- Casing is set "in the dark".
- Formation top is generally picked from drill cuttings. Difficult to selectively stimulate producing intervals (however, open-hole packers available are used sometimes).
- Wellbore may require periodic cleanout. Formation collapse is more common.
- Requires more drilling rig time when completed.
- Subsequent cased completion will restrict wellbore and completion string diameter.

Cased Hole Completion

- In the cased and perforated completion, casing is set through the producing formation and cemented. The casing is then perforated to provide communication between' the wellbore and formation



Cased Hole Advantages

- Ease of selective completion and work overoperation in the producing intervals.
- Can effectively control gas and water producing by. Selectivelyperforating and isolating.
- Can effectively control and monitor zonal fluid production. Permits multiple completions,
- Can "stimulate selectively, DSTs, logs and formation samples provide information on casing setting depth.
- Can be adopted to sand control, both as prepack gravel pack or conventional gravel pack.

Cased Hole Disadvantages

- Long interval perforation can be expensive.
- Effective wellbore diameter and productivity .may be reduced.
- Good cement job through production intervals required.
- More expensive than open-hole.

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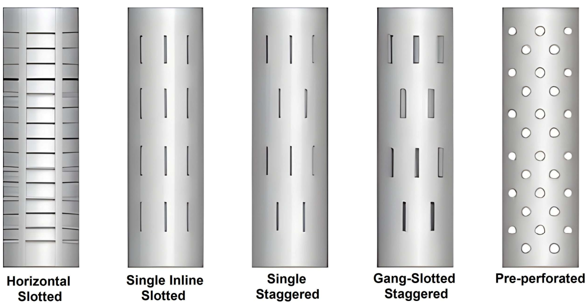
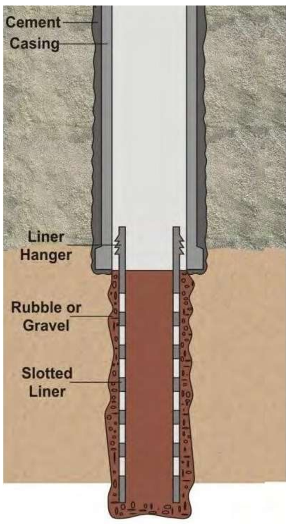
Liner Completion

- In a liner completion casing run is to the top of the pay and a liner is set across the producing formation.





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PETROLEUM PRODUCTION ENGINEERING

MODULE-6

RESERVOIR CONSIDERATIONS

LINER COMPLETIONS

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Liner Completion

- The liner may be or may not be cemented
- The cemented liner is perforated whereas un-cemented liner is slotted.
- This type of completion is primarily applied to sand control, but could be used to control a sloughing formation.
- Completions using cemented liners to reduce casing costs are essentially considered cased and perforated completions.
- The advantages and disadvantages of the un-cemented liner completion are the same as the open-hole completion.

Un-cemented Liner Completion: Advantages

- Entire pay zone open to the wellbore.
- No perforation cost.
- Log interpretation is not critical.
- Adaptable to special sand control methods.
- Cleanout problem is avoided in screen liner.
- Wire wrapped screens can be placed later.

Un-cemented Liner Completion: Disadvantage

- Excessive water or gas is difficult to control.
- Casing is set before pay zones are drilled and logged.
- Selective stimulation is not possible.

Cemented Liner Completion: Advantage

- Formation damage is minimized.
- Selective stimulation is possible in cemented liner.
- Perforation expense is avoided in screen liner.
- Cleanout problem is avoided in screen liner.

Cemented Liner Completion: Advantage

- Diameter across the pay is minimized
- Good quality cementation is difficult in cemented liner.
- Log interpretation is critical.
- Perforating, cementing and rig time incurs additional costs.

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MODULE-7

RESERVOIR CONSIDERATIONS

INFLOW PERFORMANCE AND PRODUCTIVITY INDEX

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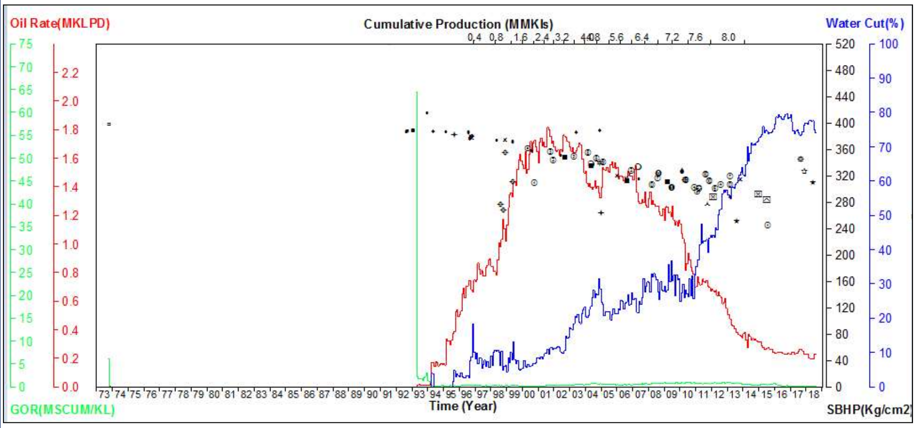
Periodic Production Test

- Production tests are done routinely to physically measure oil, gas and water produced by a particular well with a fixed bean or chalk size.
- This test provide periodic physical evidence of well conditions.
- Unexpected changes, such as extraneous water or gas production may signal well or reservoir problems.
- Abnormal production declines may mean artificial lift problems, sand fill up in the casing, scale buildup in the perforations, etc.

Performance Plot



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Productivity or Deliverability Test (Bean study)

- It involves a physical or empirical determination of produced fluid flow vs. pressure drawdown, ΔP.
- Results are used to determine optimum production rate/fluid withdrawal & to select completion method & production facility.
- This is a successfully applied to non-darcy, below the bubble point flow conditions, even though fluid properties and relative permeabilities are not constant around the wellbore.
- It does not permit calculation of formation permeability or the degree of abnormal flow restriction (formation damage) near the wellbore.

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Productivity or Deliverability Test (Bean study)

- Deliverability tests represent stabilized producing conditions.
- It involves the measurement of bottomhole static and flowing pressure, as well as fluid rates produced to the surface.
- Commonly used deliverability tests for oil wells may be classified as:
  - Productivity index
  - Inflow performance

Productivity or Deliverability Test (Bean study)

- In an oil well the productivity index test (or indicator diagram study) is the simplest form of deliverability. It is define by the equation:

$$\mathrm{PI} = \mathrm{J} = \frac{\mathrm{Q_o}}{\mathrm{P_s} - \mathrm{P_{wf}}} = \frac{\mathrm{Q_o}}{\Delta \mathrm{P}}$$

- Where,

  Q = Flow rate, m^3/day (or bbl/day)

  P_s = Shut in pressure (kg/cm^2 or psi)

  P_{wf} = Flowing reservoir pressure (kg/cm^2 or psi)

  ΔP = Pressure draw down (kg/cm^2 or psi)

Indicator Diagram Study

- It consists of measurement of shut-in bottom hole pressure (SBHP); and, at one stabilized producing conditions which includes, measurement of the flowing bottom bole pressure and the corresponding rate of liquids produced to the surface.
- The equation of the indicator line is

$$Q_o = J\left(P_s - P_{wf}\right)^n = J(\Delta P)^n$$

- Where, 'J' and 'n' are indices.
- 'J' is the productivity index and its unit is m^3/day/kg/cm^2 or bbl/day/psi
- 'n' (is not less than 1).

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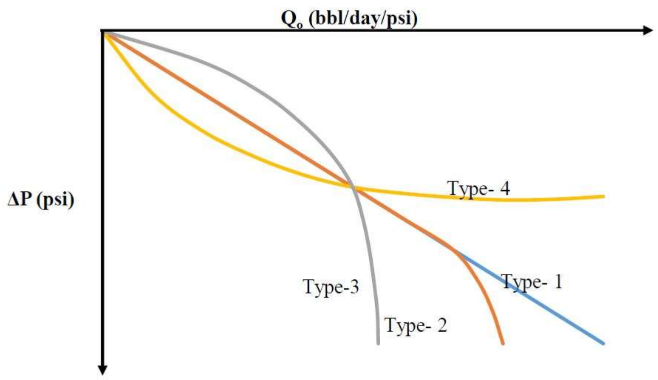
Indicator Diagram Study

- There are different types of indicator lines are obtained depending upon the value of 'n'



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Indicator Diagram Study

- Type 1: When n=1; the indicator curve is a straight line and passes through the origin. It represents the case of a steady, state single-phase fluid flow that follows the linear law of filtration (Darcy's Law).
- Type-2: When n<1, the shape of the indicator diagram is straight initially but as reservoir pressure goes below saturation pressure or bubble point pressure (Pb) curve becomes convex in shape in relation to the q-axis. The curving of the indicator line indicates the deviation from the linear low of filtration in the bottom-hole zone. This type of indicator line may be due to the appearance of a two-phase flow. From this point of divergence in the straight line saturation pressure can be approximately evaluated.

Indicator Diagram Study

- Type 3: When n<1, the shape of the curve is convex to q axis, since beginning, due to turbulent flow. When the recovery mechanism of the reservoir is other than water drive (i.e., solution gas drive or gravity drive), the indicator line of this type is observed.
- Type 4: When n>1; when the production rates and flowing bottom hole pressure are not stabilized. The shape of the curve is concave in relation to q axis. This may be obtained due to errors in measuring flowing bottom-hole pressure and oil rates. Such indicator line results from unsteady flow in the formation.

Problem 1

- A productivity test was conducted on a well. The test results indicate that the well is capable of producing at a stabilized flow rate of 110 STB/day and a bottom-hole flowing pressure of 900 psi. After shutting the well for 24 hours, the bottom-hole pressure reached a static value of 1,300 psi.
- Calculate:
  - Productivity index
  - AOF
  - Oil flow rate at a bottom-hole flowing pressure of 600 psi
  - Wellbore flowing pressure required to produce 250 STB/day

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Solution

- PI = 0.275 STB/psi
- AOF = 375.5 STB/day
- 192.5 STB/day
- 390.9 psi

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Problem 2

- A well is produced from a saturated oil reservoir with an average reservoir pressure of 3,000 psig. Stabilized flow test data indicate that the well is capable of producing 400 STB/day at a bottom-hole flowing pressure of 2,580 psig. (Assume a constant J)

a) Oil flow rate at P_{wf} = 1950 psig

b) Productivity Index

c) AOF

d) Plot the IPR curve

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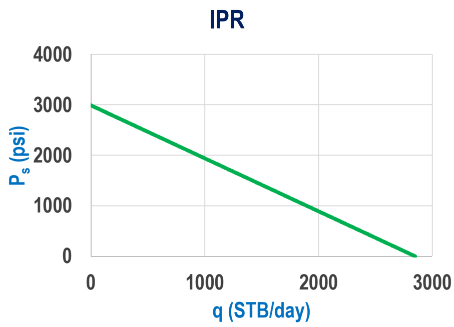
Solution

a) Oil flow rate (*at P_{wf} = 1950 psig*): 1000 STB/day

b) Productivity index: 0.952 STB/day/psi

c) AOF: 2857.14 STB/day



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Test Your Understanding

- Discuss about the role of tubulars and packers in petroleum production from subsurface formations. 15 Marks
- Discuss about the various types of completion along with their schematic diagrams and their purpose for the oil and gas production. 15 Marks
- Discuss the advantages and disadvantages of the various types of completions used in oil and gas industry. 15 Marks

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Note: Values will be changed for exam

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